

The economics of architecture

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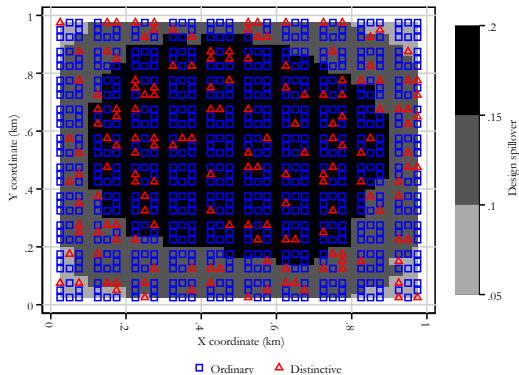
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Model

- ▶ **Geography:** Neighbourhood with J parcels and one developer per parcel
 - ▶ Each parcel has developable land $\bar{K}_i$, can be developed with one building
 - ▶ Embedded in wider city
- ▶ **Residents:** Choose a **building** (either distinctive or not) or the rest of the city
 - ▶ Standard Cobb-Douglas preferences for housing and floor space
 - ▶ Amenity B_i includes exogenous component and **design spillovers** D_i
 - ▶ Take the standard market potential form (normalized to range between 0 and 1)
 - ▶ Idiosyncratic preferences (Fréchet) for submarkets
 - ▶ **Downward-sloping demand** within submarkets: Distinctive, ordinary, rest of city
- ▶ **Developers:** Choose building height h_i and **design** d
 - ▶ Face design-specific construction cost and height elasticity of cost θ
 - ▶ Heterogeneity in design cost generates **upward-sloping aggregate supply**
 - ▶ May receive subsidy t_i^d and face design regulation G_i

Equilibrium

Intuition and quantification



- ▶ Developers develop distinctive buildings **where it is profit-maximizing**
- ▶ **Marginal parcel** is where the return to distinctive design falls below the cost
- ▶ Exposure to distinctive design **spillover translates into higher prices**
- ▶ **Need moments to calibrate the model**
 - ▶ Internal design premium
 - ▶ External design premium
 - ▶ Taste dispersion

	(1)	(2)	(3)
	Premium	Premium	Premium
Commercial	0.095*** (0.02)	0.101*** (0.02)	0.146*** (0.02)
Residential	0.077*** (0.02)	0.099** (0.04)	0.164*** (0.04)
Conservation Area		-0.02 (0.03)	
Listed Building		-0.08* (0.05)	
Award (pure)		-0.05 (0.04)	
Historic Style		0.010 (0.05)	
Weighted	No	No	Yes
Sample	All	All	Cohort control
r2	0.4	0.4	0.7
N	68	68	17

- ▶ Regression-based average
 - ▶ We weight observation by publication-year adjusted citation count
 - ▶ Most credible estimates are from studies that identify the distinctiveness effect within cohorts
 - ▶ **Internal design premium $\approx 15\%$**

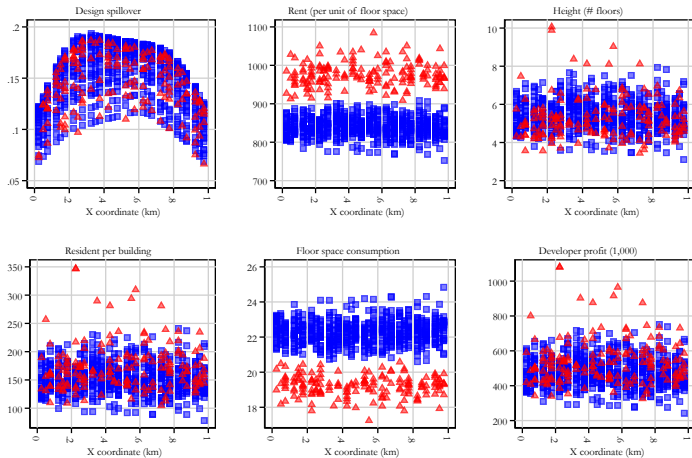
Design preferences

- ▶ **Internet survey** asking respondents to rate 10 buildings on a 0-100 scale
 - ▶ Also obtained ratings by "Starchitect" Stefano Boeri (Bosco Verticale, Milan)
- ▶ Structural estimation of **Fréchet shift and share parameters** using GMM: $\epsilon = 4$
 - ▶ Buildings rated highly by Starchitect also appreciated by respondents ($\rho = 0.8$)

Group	Mean rating	Variance	Shift A_b	Shape ϵ
All respondents, all buildings	52.72	480.88	42.73	4.04
People similar to Starchitect	51.34	416.81	39.53	4.06
People different from Starchitect	52.99	586.57	42.16	3.79
Buildings ranked high by Starchitect	65.63	486.86	55.89	4.69
Buildings ranked low by Starchitect	39.80	474.90	32.39	3.40

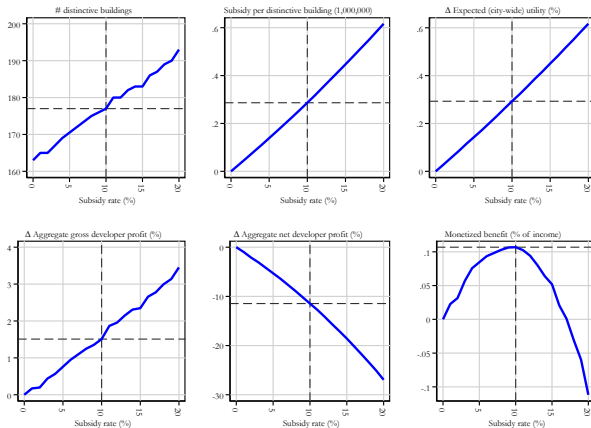
Notes: This table reports GMM estimates based on matching the first and second moments of the building-level rating distributions to those implied by a Fréchet model. The shape parameter ϵ governs preference heterogeneity, while A_b reflects the building-specific location shift. Groups are defined based on respondent-level or Starchitect-level ranking similarity.

Neighbourhood structure



- ▶ Distinctive design premium
 - ▶ Matched to data
- ▶ Affects floorspace consumption
 - ▶ Higher price $\Rightarrow$ lower consumption
- ▶ Design spillover is greatest in the centre
 - ▶ Affects prices (effect barely visible)

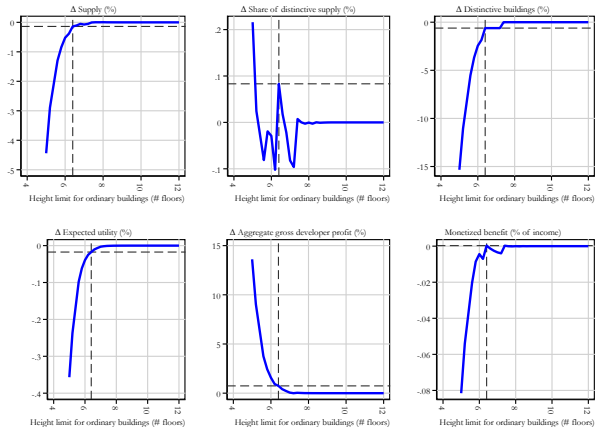
Pigovian subsidy



- ▶ Increases
 - ▶ # distinctive buildings
 - ▶ Expected utility
 - ▶ Developer gross profits
- ▶ Reduces net profits after taxes
 - ▶ Convex shape
- ▶ Hump-shaped welfare effect
 - ▶ Monetized utility effect, developer profits, and tax cost
 - ▶ Downward-sloping demand $\Rightarrow$ concavity
- ▶ **Optimal subsidy:**
 - ▶ 10% of construction cost
 - ▶ $\approx \frac{1}{3}$ of distinctive design cost

Subsidy gradients

FAR incentives



- ▶ **FAR incentives** only work under **constraints** for ordinary buildings
 - ▶ Introduce height limit for ordinary buildings
 - ▶ No limit for distinctive buildings
- ▶ Generally, binding **height limits depreciate welfare**
 - ▶ Non-binding height limits do not affect distinctive supply
- ▶ **Knife-edge welfare gain**
 - ▶ At 6.4-floor height limit
 - ▶ hard to hit in practice
 - ▶ Tighter constraints depreciate aggregate welfare and transfer welfare from workers to landowners

Super developer

- ▶ **Coordination via super-developer:**
 - ▶ Have an incentive to internalize spillovers
 - ▶ Also have an incentive to maximize monopoly rent from scarce, distinctive design...
- ▶ **Small positive effect on distinctive supply** compared to myopic developer
 - ▶ A strategic developer ignoring spillovers would supply much less distinctive design
 - ▶ Monopoly quantity setting partially offsets the incentive to internalize spillovers

Scenario comparison	Dist. build.	Dist. supply	Supply	Rents	Expected utility	Pop	Profits
Myopic → Strategic	+3.1%	+0.2%	-0.31%	+0.44%	+0.01%	+0.06%	+0.06%
Strategic → Strat. w/o spillover	-13.7%	-3.3%	+0.34%	-0.43%	-0.02%	-0.13%	-0.12%

Notes: The table reports equilibrium outcomes under three super-developer scenarios using the greedy algorithm. Under the *fully-myopic* scenario the super developer makes sequential parcel choices based on fixed rents and historical spillover. The *fully-strategic* developer is a monopolist who anticipates current equilibrium spillovers. In the *strategic without spillover* scenario, the developer is a monopolist with given spillovers.

Workers: Design spillovers and local amenity

- ▶ Local amenity:
 - ▶ $B_i^d = b_i^d \cdot \exp[\beta D_i]$
- ▶ Design spillover:
 - ▶
$$D_i = \frac{\sum_{z \neq i} \omega_{iz} \mathbf{1}(\tilde{d}_z = \text{distinctive})}{\max_i \left(\sum_{z \neq i} \omega_{iz} \right)}$$
- ▶ Spatial weight:
 - ▶ $\omega_{iz} = (\bar{K}_z h_z) \exp(-\tau \mathcal{D}_{iz})$
- ▶ Interpretation:
 - ▶ Nearby distinctive buildings raise local amenities and rents
 - ▶ Larger nearby buildings generate stronger spillovers
 - ▶ Spillovers decay with distance
- ▶ Coordination problem:
 - ▶ Developers ignore benefits to neighbouring parcels

Developers: Technology and profits

- ▶ One developer per parcel i
 - ▶ Chooses building design $d \in \{\text{distinctive}, \text{ordinary}\}$
 - ▶ Chooses height h_i^d on parcel land $\bar{K}_i$
- ▶ Revenue:
 - ▶ $Q_i^d h_i^d \bar{K}_i$
 - ▶ Q_i^d is the unit price of floor space
- ▶ Construction cost:
 - ▶ $\bar{C}(h_i^d)^{1+\theta} \exp(\delta_i^d - t_i^d) \bar{K}_i$
 - ▶ θ governs how construction costs rise with height
 - ▶ δ_i^d is a design-specific cost shifter
 - ▶ t_i^d is a construction-cost subsidy
- ▶ Profit:
 - ▶ $\pi_i^d = \left[Q_i^d h_i^d - \bar{C}(h_i^d)^{1+\theta} e^{\delta_i^d - t_i^d} \right] \bar{K}_i$

Developers: Height and design choice

- ▶ Profit-maximising height:

- ▶
$$h_i^{d*} = \left(\frac{Q_i^d}{(1+\theta)\bar{c}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$$

- ▶ Height regulation:

- ▶ $\tilde{h}_i^d = \min(h_i^{d*}, \bar{h}_i^d)$

- ▶ $\bar{h}_i^d$ is the design-specific height limit

- ▶ Design choice:

- ▶ If $G_i = \text{distinctive}$: planner requires distinctive design

- ▶ Else: developer chooses

- ▶ $\tilde{d}_i = \arg \max_{d \in \mathcal{D}} \pi_i^d$

- ▶ Floor-space supply:

- ▶ $H_i^d = \tilde{h}_i^d \mathbf{1}(\tilde{d}_i = d) \bar{K}_i$

- ▶ Key mechanism:

- ▶ Developers internalize own rents, but not spillovers on neighbouring parcels

Welfare

- ▶ Two welfare components:
 - ▶ Worker utility: $\mathbb{E}[U]\bar{N}$
 - ▶ Developer profits:
 - ▶ $\Pi = \sum_{i,d} \pi_i^d \mathbb{1}(\tilde{d}_i = d) + \bar{\pi}^{outside}$
- ▶ Expected utility:
 - ▶ $\mathbb{E}[U] = \Gamma\left(\frac{\varepsilon-1}{\varepsilon}\right) \left(\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon\right)^{1/\varepsilon}$
- ▶ Welfare metric:
 - ▶ Utility changes converted into income-equivalent units
 - ▶ Cobb-Douglas structure implies utility scales proportionally with income
- ▶ Policy evaluation:
 - ▶ $\Delta W = \frac{\mathbb{E}[U]^1 - \mathbb{E}[U]^0}{\mathbb{E}[U]^0} \bar{w} \bar{N} + \Delta \Pi$
- ▶ Subsidy cost:
 - ▶ $S = \sum_{i,d} \mathbb{1}(\tilde{d}_i = d) \bar{C}(h_i^d)^{1+\theta} \bar{K}_i \left[e^{\delta_i^d} - e^{\delta_i^d - t_i^d} \right]$

Equilibrium: Spatial equilibrium conditions

- ▶ Workers are mobile:
 - ▶ Across neighbourhoods via outside option (constant utility)
 - ▶ Across submarkets within neighbourhood (constant expected utility)
- ▶ Developers are competitive:
 - ▶ Take rents and spillovers as given
 - ▶ Choose profit-maximising design and height
- ▶ Floor-space markets clear at parcel level:
 - ▶ $f_i^d N_i^d = H_i^d$
- ▶ Submarket populations are consistent with discrete choice:
 - ▶ $\mu^d = \frac{1}{N} \sum_i N_i^d$
 - ▶ $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathcal{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Spatial equilibrium:
 - ▶ Rents adjust such that demand equals supply across submarkets

Equilibrium: Recursive structure

- ▶ Conditional on:
 - ▶ $\{\tilde{d}_i, Q_i^d, \tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- ▶ Workers:
 - ▶ Housing demand: $f_i^d = (1 - \alpha) \frac{\bar{w}}{Q_i^d}$
 - ▶ Choice probabilities: $\mu^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Developers:
 - ▶ Optimal height: $h_i^{d*} = \left(\frac{Q_i^d}{(1+\theta)\bar{c}e^{\delta_i^d - t_i^d}} \right)^{1/\theta}$
 - ▶ Feasible height: $\tilde{h}_i^d = \min(h_i^{d*}, \bar{h}_i^d)$
- ▶ Implied equilibrium objects:
 - ▶ Spillovers D_i , amenities B_i^d , supply H_i^d
 - ▶ Population N_i^d , utility $\mathbb{E}[U]$, profits Π

Equilibrium: Fixed-point problem

- ▶ Unknown equilibrium objects:
 - ▶ Design choices $\tilde{d}_i$
 - ▶ Rents Q_i^d
 - ▶ Submarket utilities $\{\tilde{V}^{distinctive}, \tilde{V}^{ordinary}\}$
- ▶ Submarket utilities pinned down by:
 - ▶ Population consistency: $\mu^d = \frac{1}{N} \sum_i N_i^d = \frac{(\tilde{V}^d)^\varepsilon}{\sum_{u \in \mathbf{C}} (\tilde{V}^u)^\varepsilon}$
- ▶ Design choices pinned down by:
 - ▶ Profit maximisation: $\tilde{d}_i = \arg \max_{d \in \mathbf{D}} \pi_i^d$
- ▶ Rents pinned down by:
 - ▶ Parcel-level market clearing: $f_i^d N_i^d = H_i^d$
- ▶ Numerical solution:
 - ▶ Fixed point because spillovers affect both rents and design choices
 - ▶ Avoid multiplicity via convexification (heterogenous design cost)

Survey Examples I



40 (23)



48 (22)



29 (19)



39 (23)



61 (25)



75 (21)



72 (19)



76 (19)



50 (24)



42 (21)



37 (21)



55 (25)



73 (19)



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34 (21)



46 (24)



35 (20)



62 (22)



66 (24)



57 (23)

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Survey: Fréchet taste heterogeneity

- ▶ Survey data:
 - ▶ Respondents rate buildings on a 0–100 scale
 - ▶ Ratings interpreted as latent taste draws a_{ib}
- ▶ Structural assumption:
 - ▶ a_{ib} follows a Fréchet distribution
 - ▶ $F(a) = \exp(-(a/A_b)^{-\varepsilon})$
- ▶ Parameters:
 - ▶ A_b : building-specific attractiveness
 - ▶ ε : taste heterogeneity across respondents
- ▶ Theoretical moments:
 - ▶ $\mathbb{E}[a_b] = A_b \Gamma(1 - \frac{1}{\varepsilon})$
 - ▶ $\text{Var}(a_b) = A_b^2 \left[\Gamma(1 - \frac{2}{\varepsilon}) - \Gamma(1 - \frac{1}{\varepsilon})^2 \right]$

GMM Estimation

► Empirical moments:

$$\text{► } \hat{\mu}_b = \frac{1}{n_b} \sum_{i=1}^{n_b} a_{ib}$$

$$\text{► } \hat{\sigma}_b^2 = \frac{1}{n_b-1} \sum_{i=1}^{n_b} (a_{ib} - \hat{\mu}_b)^2$$

► Moment conditions:

$$\text{► } g_b^\mu(\theta) = \hat{\mu}_b - A_b \Gamma \left(1 - \frac{1}{\varepsilon}\right) = 0$$

$$\text{► } g_b^\sigma(\theta) = \hat{\sigma}_b^2 - A_b^2 \left[\Gamma \left(1 - \frac{2}{\varepsilon}\right) - \Gamma \left(1 - \frac{1}{\varepsilon}\right)^2 \right] = 0$$

► Identification:

► 20 buildings $\Rightarrow$ 40 empirical moments

► Mean and variance for each building

► 21 parameters (ε and 1 A_b per building) $\Rightarrow$ overidentified system

► Estimation method: One-step GMM with identity weighting matrix

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SMM: Calibration targets

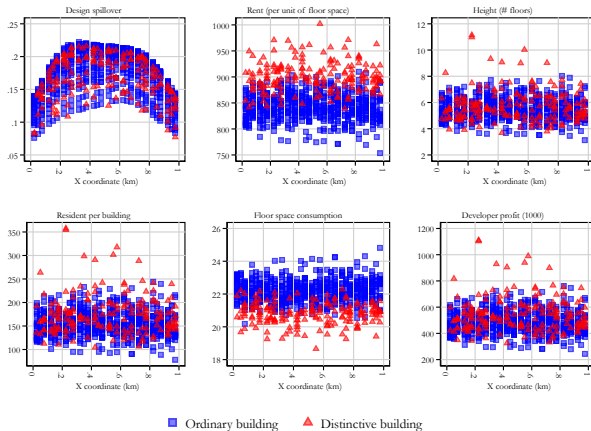
- ▶ Parameters to calibrate:
 - ▶ Preference for distinctive design: $A^{d=\text{distinctive}}$
 - ▶ Average distinctive design cost: $\bar{\delta}^{d=\text{distinctive}}$
- ▶ Distinctive design cost: Common component plus shock from normal distribution
 - ▶ $\delta_i^{d=\text{distinctive}} = \bar{\delta}^{d=\text{distinctive}} \left(1 + \sigma \frac{\tilde{\delta}_i}{\bar{\delta}^{d=\text{distinctive}}} \right)$
- ▶ Empirical target moments:
 - ▶ Internal design premium: $\hat{a} = 15\%$
 - ▶ Share of distinctive buildings: $\hat{s} = 20\%$
- ▶ SMM objective:
 - ▶ Match model-implied premium and distinctive share
 - ▶ Internal premium identifies preferences
 - ▶ Distinctive share identifies costs

SMM: Numerical procedure

- ▶ Inner loop:
 - ▶ Solve developer design choices and heights
 - ▶ Iterate until no parcel switches design
- ▶ Middle loop:
 - ▶ Update utilities and population shares
 - ▶ Clear housing markets across submarkets
- ▶ Outer loop:
 - ▶ Update structural parameters: $A^{d=\text{distinctive}}$,
 - ▶ $\bar{\delta}^{d=\text{distinctive}}$
- ▶ Numerical objective:
 - ▶ Minimize weighted distance between:
 - ▶ Simulated and empirical internal premium
 - ▶ Simulated and empirical distinctive share
- ▶ Main calibration:
 - ▶ $A^{d=\text{distinctive}} = 0.772$, $\bar{\delta}^{d=\text{distinctive}} = 0.219$

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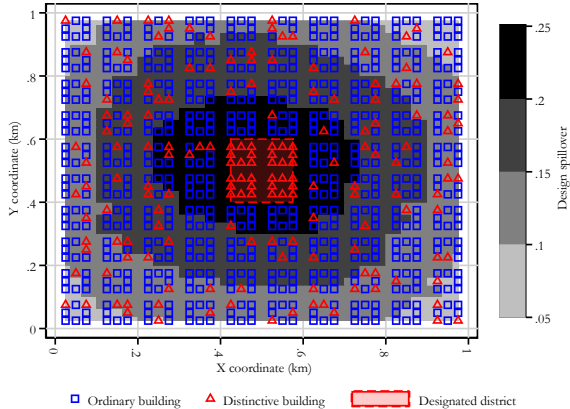
Subsidy Gradients



- ▶ Effects across subsidy levels:
 - ▶ Distinctive building share rises smoothly
 - ▶ Rents increase, but at diminishing rate
- ▶ Margins of adjustment:
 - ▶ Utility gain from spillovers
 - ▶ Developer profits vs. tax burden
- ▶ Welfare profile:
 - ▶ Concave, consistent with downward-sloping demand
 - ▶ Peak defines Pigovian optimum

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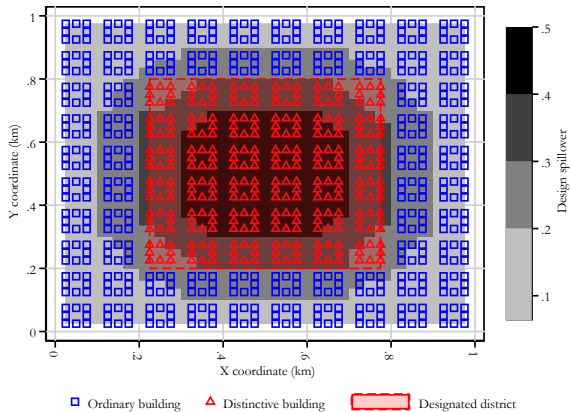
Mandatory District Map



- Visualizes the location of the mandatory design district
- Shows spatial clustering of distinctive buildings
- Useful for interpreting spillover and coordination effects

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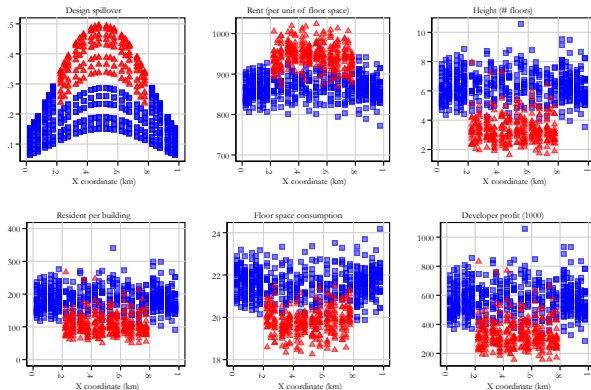
Mandatory District Map



- ▶ All distinctive buildings now in the district
- ▶ Many more distinctive buildings
- ▶ Profitable distinctive buildings **crowded out**

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Mandatory District: Large-Scale Scenario

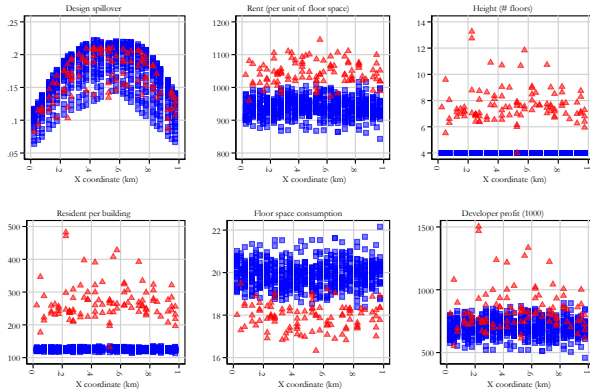


■ Ordinary building ▲ Distinctive building

- ▶ When the mandatory district is large:
 - ▶ Distinctive share saturates
 - ▶ Returns diminish, congestion effects rise
- ▶ Welfare impact turns negative
 - ▶ Utility and profits decline
- ▶ Policy trade-off:
 - ▶ Small districts solve coordination
 - ▶ Large ones impose inefficiency

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FAR incentives: Gradients



■ Ordinary building ▲ Distinctive building

- ▶ With binding height constraints on ordinary buildings
 - ▶ Heights of distinctive buildings increase
 - ▶ Positive distinctive design premium persists
 - ▶ Some distinctive buildings generate large developer profits

Back to FAR incentives